

2016.0 RANGE ROVER (LG), 501-20

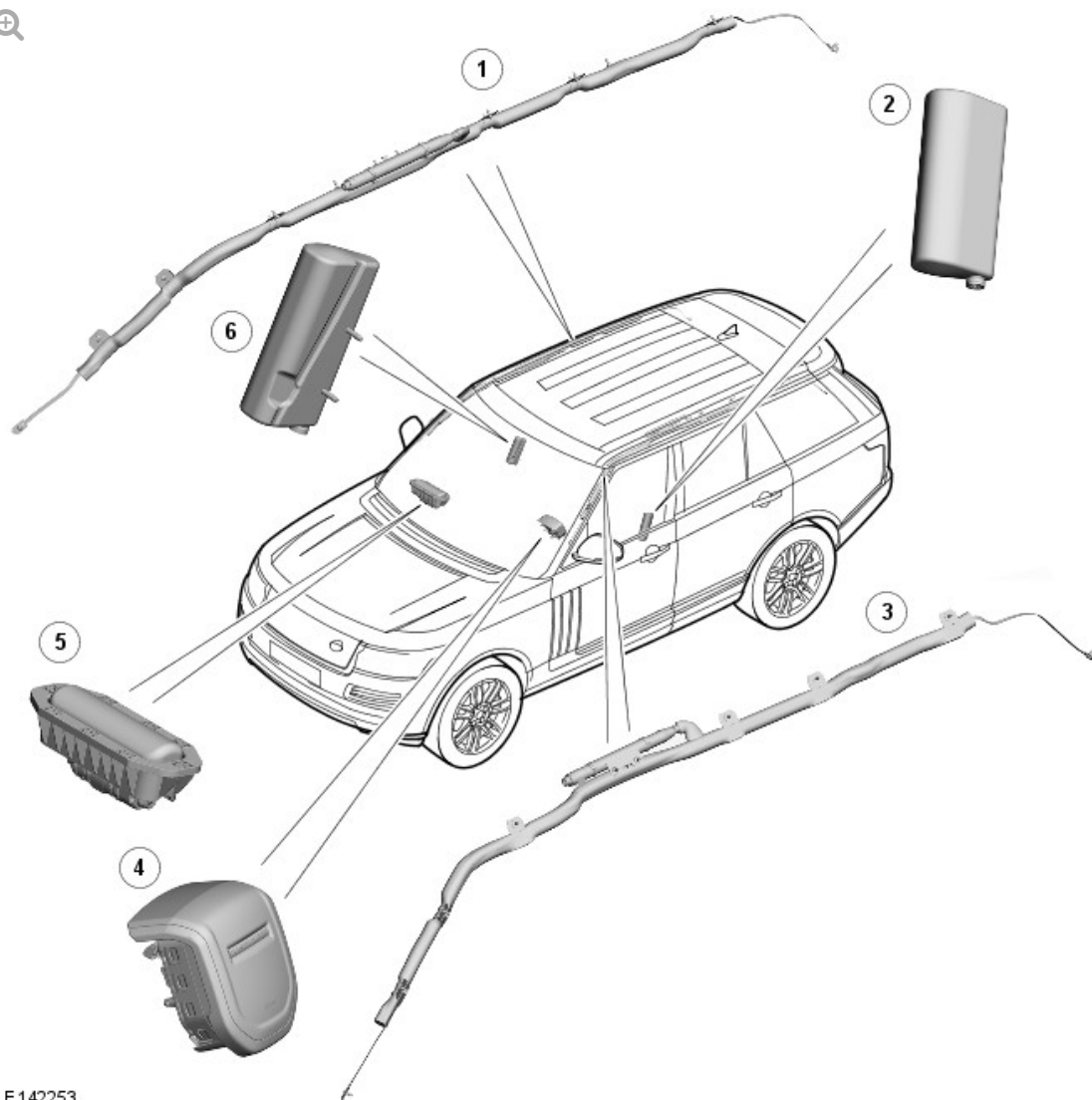
SUPPLEMENTAL RESTRAINT SYSTEM

DESCRIPTION AND OPERATION

COMPONENT LOCATION - SHEET 1 OF 3

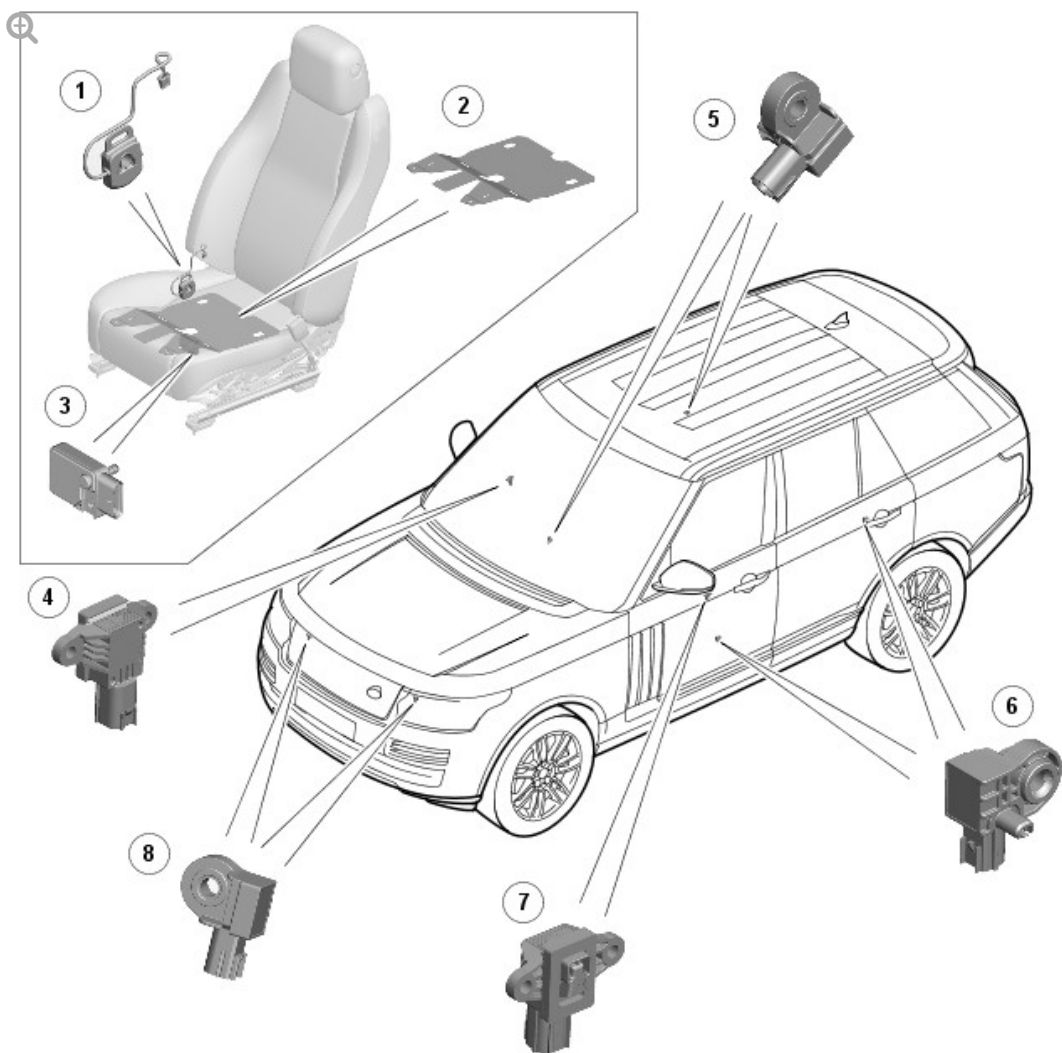
NOTE:

LHD (left-hand drive) installation shown, RHD (right-hand drive)
installation similar.



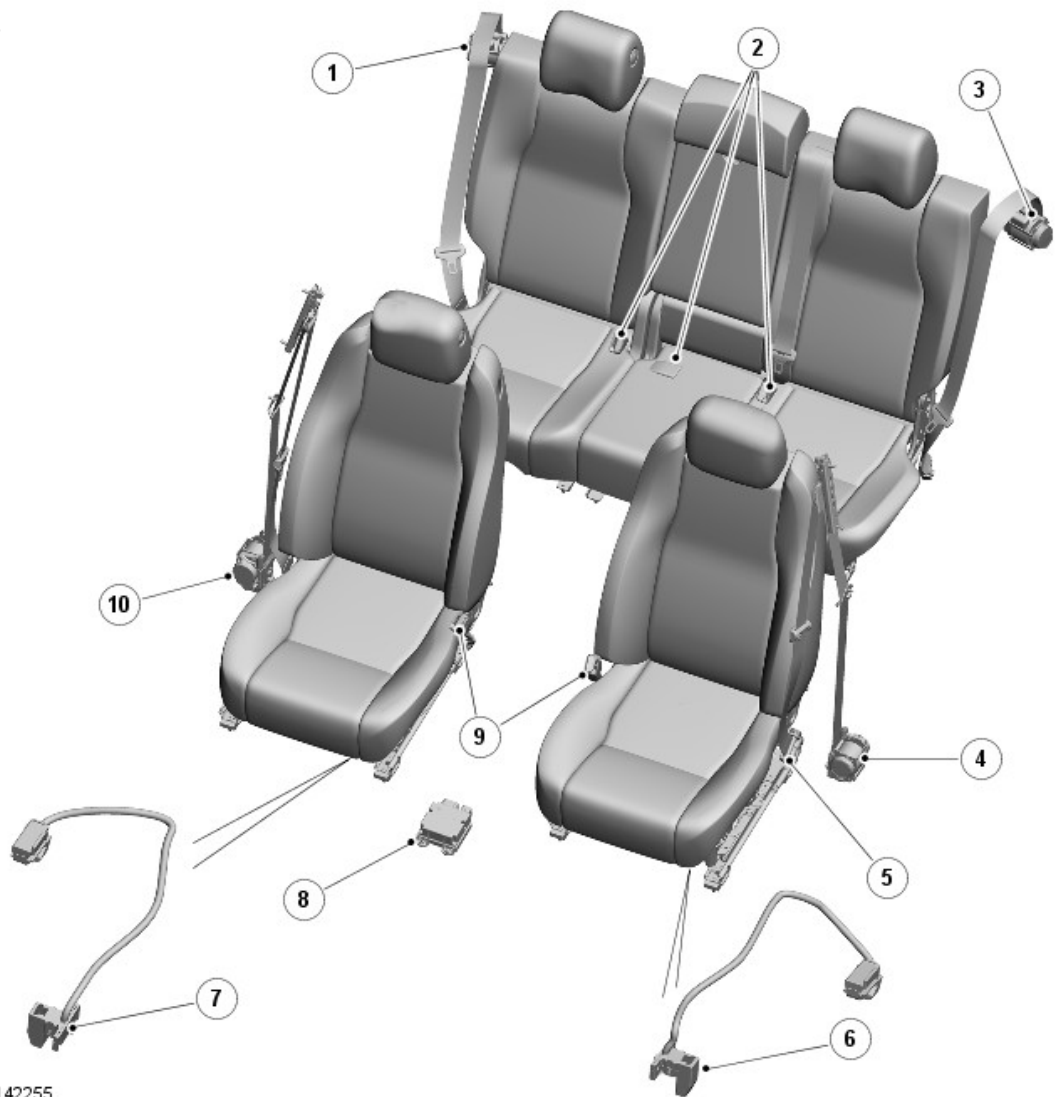
E142253

ITEM	DESCRIPTION
1	Right side air curtain
2	Driver side airbag
3	Left side air curtain
4	Driver airbag
5	Passenger airbag
6	Passenger side airbag



E142254

ITEM	DESCRIPTION
1	Seatbelt tension sensor (NAS only)
2	Pressure pad (NAS only)
3	Occupant classification sensor control module (NAS only)
4	Right side impact pressure sensor
5	Right side impact sensors
6	Left side impact sensors
7	Left side impact pressure sensor
8	Front impact sensors



E142255

ITEM	DESCRIPTION
1	Right rear pretensioner retractor
2	Rear seatbelt buckle switches
3	Left rear pretensioner retractor
4	Driver pretensioner retractor
5	Driver seatbelt pretensioner
6	Driver seat position sensor
7	Passenger seat position sensor (NAS only)
8	Restraints control module

9	Front seatbelt buckle switches
10	Front passenger pretensioner retractor

OVERVIEW

WARNING:

All pyrotechnic devices are dangerous. Before performing any procedures on any pyrotechnic device, read all information contained within the Standard Workshop Practices section of this manual.

For additional information, refer to: [Standard Workshop Practices](#) (100-00 General Information, Description and Operation).

The SRS (supplemental restraint system) provides additional protection for vehicle occupants in certain impact conditions by selective activation of driver and passenger airbags, side airbags, side air curtains and seatbelt pretensioners. Operation of the system is controlled by a RCM (restraints control module).

The RCM receives inputs from various sensors around the vehicle to determine which devices, if any, should be activated during an accident. The inputs include those from an occupant classification system for the front passenger seat (NAS vehicles only), front and side impact sensors and side impact pressure sensors.

On all except NAS, Australia and Japan market vehicles, the passenger airbag can be enabled and disabled using the vehicle Set-up menu in the instrument cluster. For additional information, refer to: (413-01 Instrument Cluster)

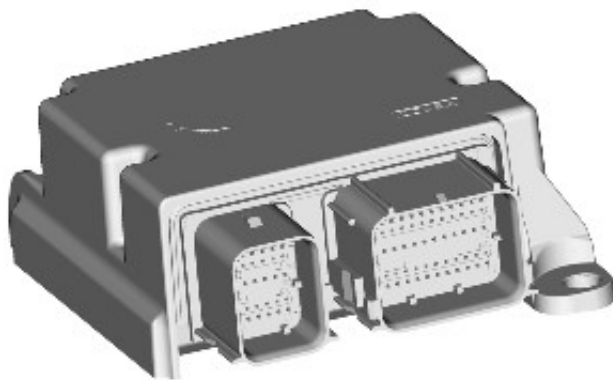
[Instrument Cluster](#) (Description and Operation),
[Instrument Cluster - SDV6 3.0L Diesel - Hybrid Electric Vehicle](#) (Description and Operation).

The activation status of the passenger airbag is given by an indicator in the front overhead console. The status of the SRS is given by an airbag warning indicator in the instrument cluster.

The airbags and air curtains feature hybrid inflators, which use a self-contained pyrotechnic device to heat a cylinder of compressed inert gas within the inflator. Above a calibrated pressure, the cylinder bursts and releases the gas into the airbag. The benefits of hybrid inflators are a cooler inflation gas, resulting in lower airbag surface temperatures, and elimination of the post-deployment effluent associated with solid pyrotechnic inflators.

DESCRIPTION

RESTRAINTS CONTROL MODULE



E135956

The RCM is installed on top of the floor tunnel, under the floor console, and controls operation of the SRS. The main functions of the RCM include:

- Crash detection and recording.
- Airbag, air curtain and pretensioner firing.
- Self test and system monitoring, with status indication via the airbag warning indicator, and non volatile storage of fault information.

- Supply of yaw rate and lateral acceleration data for the ABS (anti-lock brake system) control module.

The RCM determines which elements of the SRS are to be deployed by using two internal areas:

- Crash severity evaluation.
- Deployment handler.

Crash severity evaluation uses data from the RCM internal accelerometer, the front impact sensors and the seatbelt buckle switches. Based on this data, the RCM decides which level of airbag module deployment is required and forwards the information to the second area, the deployment handler.

The deployment handler evaluates the status of the seat position sensor(s), seatbelt buckle switches and, on NAS vehicles, the occupant classification system, before a final decision is made about which restraints should be deployed.

Data from the side impact and pressure sensors is used by the RCM in conjunction with acceleration data from the RCM internal accelerometer to make a deployment decision. The RCM processes the acceleration data and, subject to an impact being of high enough severity, decides whether the side airbag and air curtain should be deployed.

Yaw rate and lateral acceleration data for the ABS control module is derived from the internal accelerometer and transmitted on the high speed CAN (controller area network) buses.

On board testing of the airbag, air curtain and pretensioner firing circuits, the warning indicator circuits and the RCM status is performed by the RCM, together with the storing of fault codes. The impact and pressure sensors perform basic self-tests.

The RCM drives the airbag warning indicator via a high speed CAN signal. If the warning indicator fails, a fault code is recorded and a warning tone is

sounded in place of the indicator if a further fault occurs. In the event of a crash, the RCM records certain data which can be accessed via the diagnostic connector.

A safing sensor in the RCM provides confirmation of an impact to verify if airbag, air curtain and pretensioner activation is necessary. A roll-over sensor monitors the lateral attitude of the vehicle. Various firing strategies are employed by the RCM to ensure that during an accident only the appropriate airbags, air curtains and pretensioners are fired. The firing strategy used also depends on the inputs from the seatbelt buckle switches and, on NAS vehicles, the occupant classification system.

An energy reserve in the RCM ensures there is always a minimum of 150 milliseconds of stored energy available if the power supply from the ignition circuit is disrupted during a crash. The stored energy is sufficient to produce firing signals for the driver airbag, the passenger airbag and the seatbelt pretensioners.

When the ignition is switched on the RCM performs a self test and then performs cyclical monitoring of the system. If a fault is detected the RCM stores a related fault code and illuminates the airbag warning indicator. The faults can be retrieved by Land Rover approved diagnostic equipment. If a fault that could cause a false fire signal is detected, the RCM disables the respective firing circuit, and keeps it disabled during a crash event.

DRIVER AIRBAG

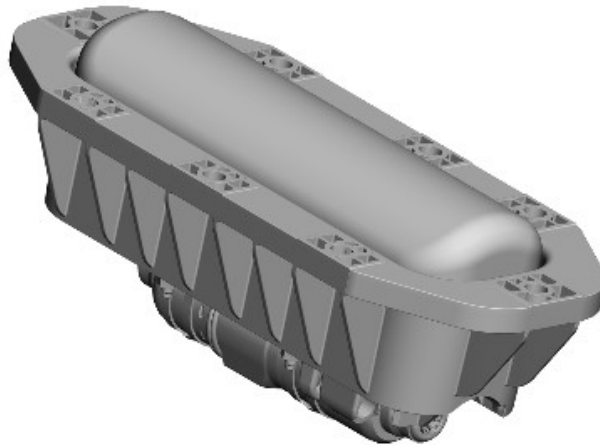


E142256

The driver airbag forms the center pad of the steering wheel. It provides head and chest protection for the driver when deployed during a front or rear impact.

Lines, molded into the inner surface of the airbag cover, provide weak points that split open in a controlled manner when the airbag deploys. The driver airbag has a two stage inflator, with separate electrical connectors on each stage.

PASSENGER AIRBAG



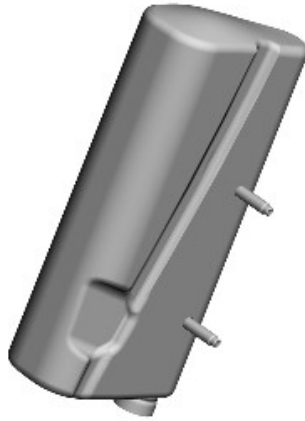
E142257

The passenger airbag is located in the instrument panel, behind the upper glovebox. It provides head and chest protection when deployed during a front or rear impact.

The top of the passenger airbag is attached to a channel chute underneath the top pad of the instrument panel. Doors in the top of the channel chute are forced open, and split the top pad, when the airbag deploys.

The passenger airbag has a two stage inflator, with separate electrical connectors on each stage.

SIDE AIRBAGS



E142258

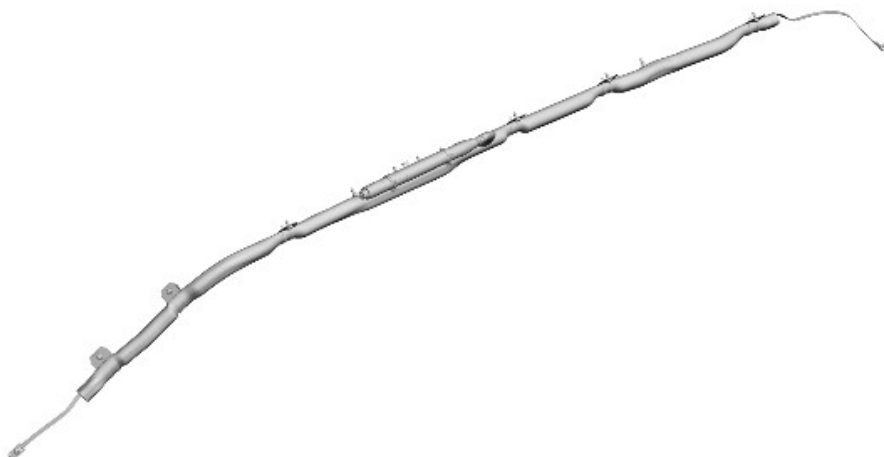
The side airbags are attached to the outside of each front seat backrest frame, under the backrest cover. They provide protection for the thorax, rib and pelvis when deployed during a side impact.

The side airbags are handed and each consists of a folded airbag and an inflator in a soft cover / wrapper. When the airbag deploys it fills within a deployment chute integrated into the seat trim, and splits open the airbag deployment seam in the backrest cover.

SIDE AIR CURTAINS

NOTE:

Standard wheelbase version shown; long wheelbase version similar.

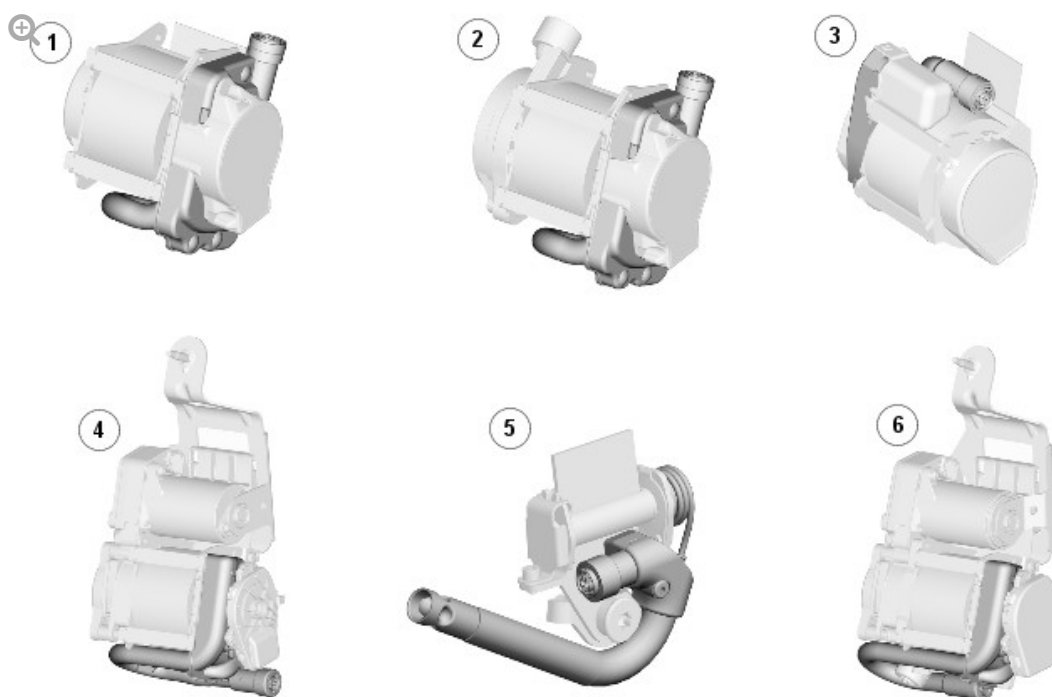


E142259

The side air curtains are installed along both sides of the car, under the edges of the headliner. They provide head protection and roll-over ejection protection for outer seat occupants, in both the front and rear of the vehicle, when deployed during a side impact.

The front and rear ends of the side air curtains are anchored to the body, to maintain tension across their lower edges of the air curtains when they deploy. Their deployment area extends between the A and C pillar trims, passing over the upper B pillar trim.

PRETENSIONERS



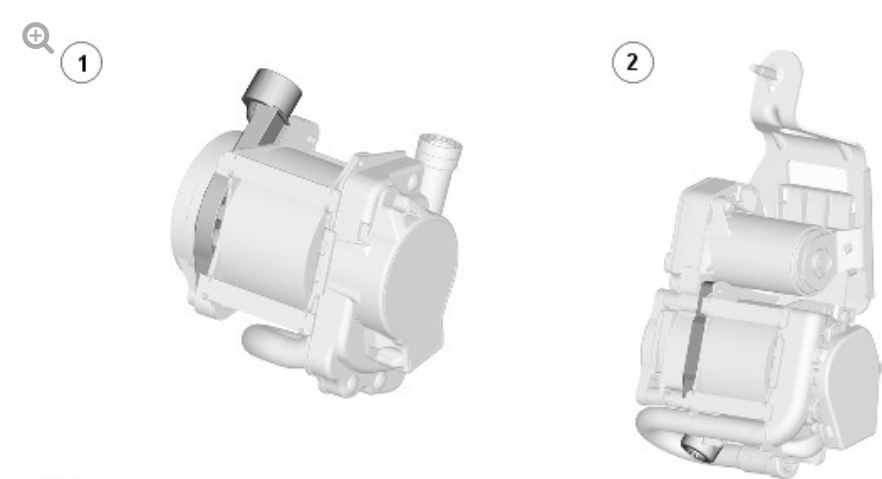
E142260

ITEM	DESCRIPTION
1	Front pretensioner retractor - non-active seatbelt (all except NAS passenger)
2	Front pretensioner retractor - non-active seatbelt (NAS passenger only)
3	Rear pretensioner retractor
4	Front pretensioner retractor - active seatbelt (all except NAS passenger)
5	Driver seatbelt pretensioner
6	Front pretensioner retractor - active seatbelt (NAS passenger)

A pyrotechnic pretensioner is integrated into each seatbelt retractor and, in some markets, the anchor point of the driver seatbelt. The pretensioners are used to tighten the seatbelts during a collision to ensure the occupants are securely held in their seats.

When operated by the RCM, the pretensioners rotate the spool holding the end of the webbing to tighten the seatbelt.

LOAD LIMITER ACTUATOR (NAS FRONT PASSENGER SEATBELT ONLY)

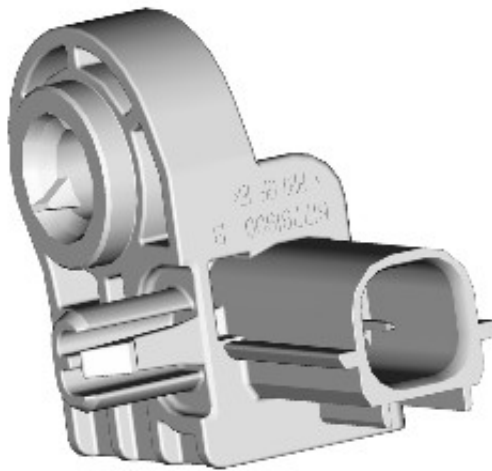


E142261

ITEM	DESCRIPTION
1	Front pretensioner retractor - non-active seatbelt
2	Front pretensioner retractor - active seatbelt

On NAS vehicles, a pyrotechnic load limiter actuator is incorporated into the retractor of the front passenger seatbelt to provide an adaptive load limiter. The adaptive load limiter has two different load limits, the standard load limit and a lower load limit. The lower load limit is designed to reduce the chest loading for smaller seat occupants during an impact, and is engaged when the RCM fires the load limiter actuator.

IMPACT SENSORS



E135952

Impact sensors are installed in the front and both sides of the vehicle. The two front impact sensors are attached to the front end carrier, behind the inboard edge of each headlamp. The side impact sensors are attached to the base of each C/D pillar and to the inner panel of each rear quarter wheel housing, adjacent to the rear door latch strikers.

The impact sensors are accelerometers that allow the RCM to detect the sudden vehicle acceleration that occurs during an impact.

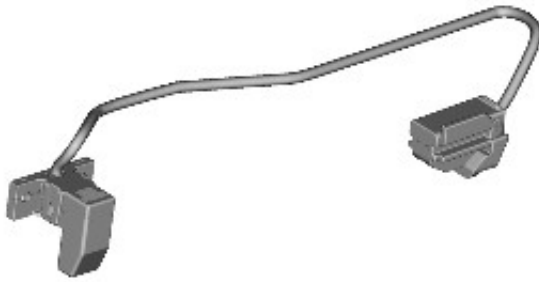
IMPACT PRESSURE SENSORS



E128623

An impact pressure sensor is installed in each front door, attached to the door closing panels. The pressure sensors allow the RCM to detect any sudden pressure pulse that occurs in the front door during a side impact.

SEAT POSITION SENSORS



E128624

The seat position sensors allow the RCM to detect when a seat is forward of a given point on the seat track. On all vehicles except NAS, a seat position sensor is installed for the driver seat only. On NAS vehicles, a seat position sensor is installed for both the driver and front passenger seats.

The seat position sensors consist of a Hall effect sensor attached to the seat frame. While the ignition is on, the RCM supplies the sensor with power, and monitors the return current. When the seat frame moves forwards, the sensor moves over the edge of the seat track, which changes the reluctance of the sensor. The change of current is detected by the RCM and used as a switching point. The switching point is when the center of the sensor is 3 ± 4 mm from the leading edge of the seat track.

OCCUPANT MONITORING

There are two types of occupant monitoring:

- In all markets except NAS, vehicles have an occupant detection sensor.
- In NAS markets, vehicles have an occupant classification system.

OCCUPANT DETECTION SENSOR



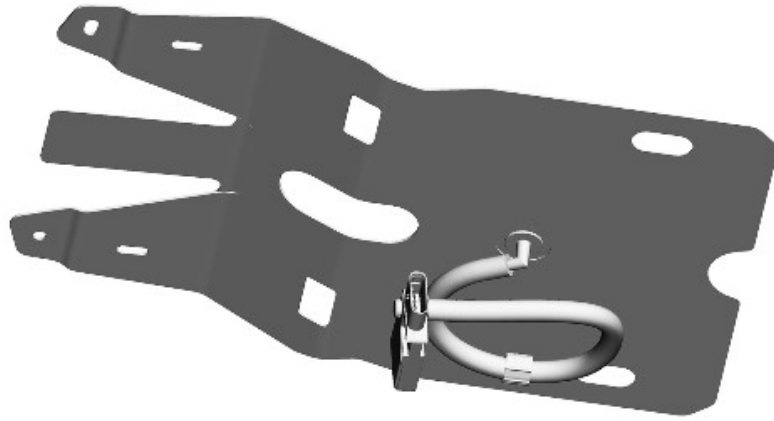
The occupant detection sensor is installed in the cushion of the front passenger seat, between the foam padding and the cover. The sensor consists of a foil contact circuit, embedded in a plastic sheet. Weight on the sensor reduces the resistance of the circuit, which is detected by the CJB (central junction box).

The occupant detection sensor has no interface with the SRS. It only provides an input to the CJB for the seatbelt reminder function.

For additional information, refer to: [Seatbelt System](#) (501-20A Seatbelt System, Description and Operation).

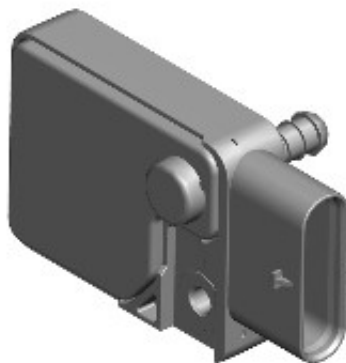
OCCUPANT CLASSIFICATION SYSTEM

Pressure Pad



E142262

Occupant Classification Sensor Control Module



E142263

Seatbelt Tension Sensor



E98178

The occupant classification system provides the RCM with the occupancy status of the front passenger seat. The RCM uses this and the seat buckle status in the evaluation of the firing strategy for the passenger airbag, the

passenger side airbag and the pretensioner retractor of the front passenger seatbelt.

The occupant classification system can determine if the front passenger seat is unoccupied, occupied by a small person, or occupied by a large person.

The occupant classification system consists of:

- An occupant classification sensor control module, installed under the front passenger seat.
- A pressure pad, installed under the cushion of the front passenger seat, with a hose connecting it to the occupant classification sensor control module.
- A seatbelt tension sensor, integrated into the anchor point of the front passenger seatbelt.

CAUTION:

The occupant classification sensor control module, hose and pressure pad form a sealed assembly and must not be disassembled. The assembly can only be replaced as part of the occupant classification system service kit. The service kit includes the seat cushion pad, because the assembly is specifically calibrated to the pad, and each pad has a unique calibration.

The pressure pad is a silicone filled bladder, with a hose connecting it to the occupant classification sensor control module. Any load on the pressure pad is detected by the occupant classification sensor control module.

The seatbelt tension sensor is a strain gauge that measures the load applied by the seatbelt anchor to the anchor bolt.

The occupant classification sensor control module supplies a reference voltage to the seatbelt tension sensor and, from the return signal, measures the load acting on the seatbelt anchor. The measurement is used to produce

a correction factor for the load acting on the pressure pad. The tightness of the seatbelt affects the load acting on the pressure pad, so without the correction factor the occupant classification sensor control module cannot derive an accurate occupancy status.

The occupant classification sensor control module translates the load readings into a seat occupancy status and transmits the result to the RCM, on the high speed CAN powertrain systems. The occupant classification sensor control module incorporates two load limits for the seat cushion: When the load is less than the lower limit, the seat is classified as unoccupied; when the load exceeds the lower limit, but is less than the upper limit, the seat is classified as occupied by a small person; when the upper limit is exceeded, the seat is classified as occupied by a large person.

The occupant classification system has four possible states as detailed in the following table:

CLASSIFICATION	SEAT STATUS	PASSENGER AIRBAG STATUS	AIRBAG INDICATOR STATUS
Empty	Empty	Disabled	Off
Occupied inhibit	The seat is occupied by a small person or a child restraint is being used	Passenger airbag and passenger side airbag operation is disabled	On
Occupied allow	The seat is occupied by a large person	Passenger airbag and passenger side airbag operation is enabled	Off
Error	-	Passenger airbag and passenger side airbag operation is disabled	On

PASSENGER AIRBAG DEACTIVATION INDICATOR



The PAD indicator is installed in the front overhead console. When appropriate, the indicator illuminates to advise front seat occupants that the passenger airbag is disabled. Operation of the indicator is controlled by the RCM connecting/ disconnecting a ground signal from the indicator. The RCM illuminates the indicator when:

- There is a fault with the passenger airbag firing circuit(s).
- The passenger airbag is disabled in the instrument cluster Set-up menu and the front passenger seat is occupied (all except NAS, Australia and Japan).
- Required by the occupant classification system (NAS only).

AIRBAG WARNING INDICATOR



E142265

Operation of the airbag warning indicator in the instrument pack is controlled by a high speed CAN powertrain systems message from the RCM. The RCM sends the signal to illuminate the airbag warning indicator if a fault is detected, and for approximately 6 seconds during the bulb check at the beginning of each ignition cycle.

OPERATION

In a collision, the force of the impact is measured by the impact sensors, impact pressure sensors and the RCM internal accelerometer. The RCM evaluates the readings to determine the impact point on the vehicle and whether the readings exceed the limits for firing any of the airbags and pretensioners. During a collision, the RCM only fires the airbags, air curtains and pretensioners if the safing function confirms that the data from the impact sensor(s) indicates an impact limit has been exceeded.

The RCM incorporates the following impact thresholds to cater for different accident scenarios:

- Front impact, pretensioners.
- Front impact, driver and passenger airbags stage 1, belt unfastened.
- Front impact, driver and passenger airbags stage 1, belt fastened.

- Front impact, driver and passenger airbags stage 2, belt unfastened.
- Front impact, driver and passenger airbags stage 2, belt fastened.
- Rear impact.
- Driver side impact.
- Passenger side impact.

FIRING STRATEGIES

The seatbelt pretensioners are fired when the pretensioner impact limit is exceeded. The RCM only fires the pretensioners if the related seatbelt is fastened.

The driver and passenger airbags are only fired in a frontal impact. If an impact exceeds a stage 1 limit, but is less than the corresponding stage 2 limit, only one inflator in each airbag is fired (stage 2 is still fired, for airbag disposal purposes, after a delay of 100 ms). If an impact exceeds the stage 2 limit, the two inflators in each airbag are fired simultaneously. When the driver seat is forward of the seat position sensor switching point, the RCM introduces a time delay between firing the two inflators of the driver airbag.

On NAS vehicles, the passenger airbag is disabled unless the front passenger seat is occupied by a large person. In all markets except NAS, Australia and Japan, the passenger airbag can be selected off in the vehicle Set-up menu of the instrument cluster.

The stage 2 inflator of the driver airbag is disabled if the driver seat is forward of the switching point of the seat position sensor.

On NAS vehicles, the load limiter actuator, in the retractor of the front passenger seatbelt, is fired if the stage 2 front impact threshold is exceeded and the passenger seat is forward of the seat position sensor switching point.

If there is a fault with a seatbelt buckle switch, the RCM assumes the related seatbelt is fastened for the pretensioner firing strategy and unfastened for the driver and passenger airbag firing strategies. If there is a fault with the

occupant classification sensor, the RCM disables the passenger airbag. If there is a fault with the passenger airbag deactivation switch, the RCM disables the passenger airbag.

If a side impact limit is exceeded, the RCM fires the side airbag and the side head airbag on that side of the vehicle. If the side impact limit on the front passenger side of the vehicle is exceeded, the RCM also evaluates the input from the occupant classification sensor, and fires the side airbag only if the front passenger seat is occupied by a large person (NAS only).

If multiple impacts occur during a crash event, after responding to the primary impact the RCM will output the appropriate fire signals in response to any further impacts if unfired units are available.

Front and Rear Impact Firing Strategy (All Except NAS)

SEATBELT STATUS			STRATEGY	
DRIVER	FRONT PASSENGER	APPLICABLE PRETENSIONER	DRIVER AIRBAG	PASSENGER AIRBAG
Fastened	-	Fired at pretensioner threshold	Fired at belt fastened threshold	-
Unfastened	-	Not fired	Fired at belt unfastened threshold	-
-	Fastened	Fired at pretensioner threshold	-	Fired at belt fastened threshold
-	Unfastened	Not fired	-	Fired at belt unfastened threshold

Front and Rear Impact Firing Strategy (NAS)

SEATBELT STATUS		PASSENGER SEAT CLASSIFICATION	STRATEGY		
DRIVER	PASSENGER		APPLICABLE PRETENSIONER	DRIVER AIRBAG	PASSENGER AIRBAG
Fastened	-	-	Fired at	Fired at	-

			pretensioner threshold	belt fastened threshold	
Unfastened	-	-	Not fired	Fired at belt unfastened threshold	-
-	Fastened	Occupied allow	Fired at pretensioner threshold	-	Fired at belt fastened threshold
-	Fastened	Occupied inhibit or empty	Fired at pretensioner threshold	-	Not fired
-	Unfastened	Occupied allow	Not fired	-	Fired at belt unfastened threshold
-	Unfastened	Occupied inhibit or empty	Not fired	-	Not fired

CRASH SIGNAL

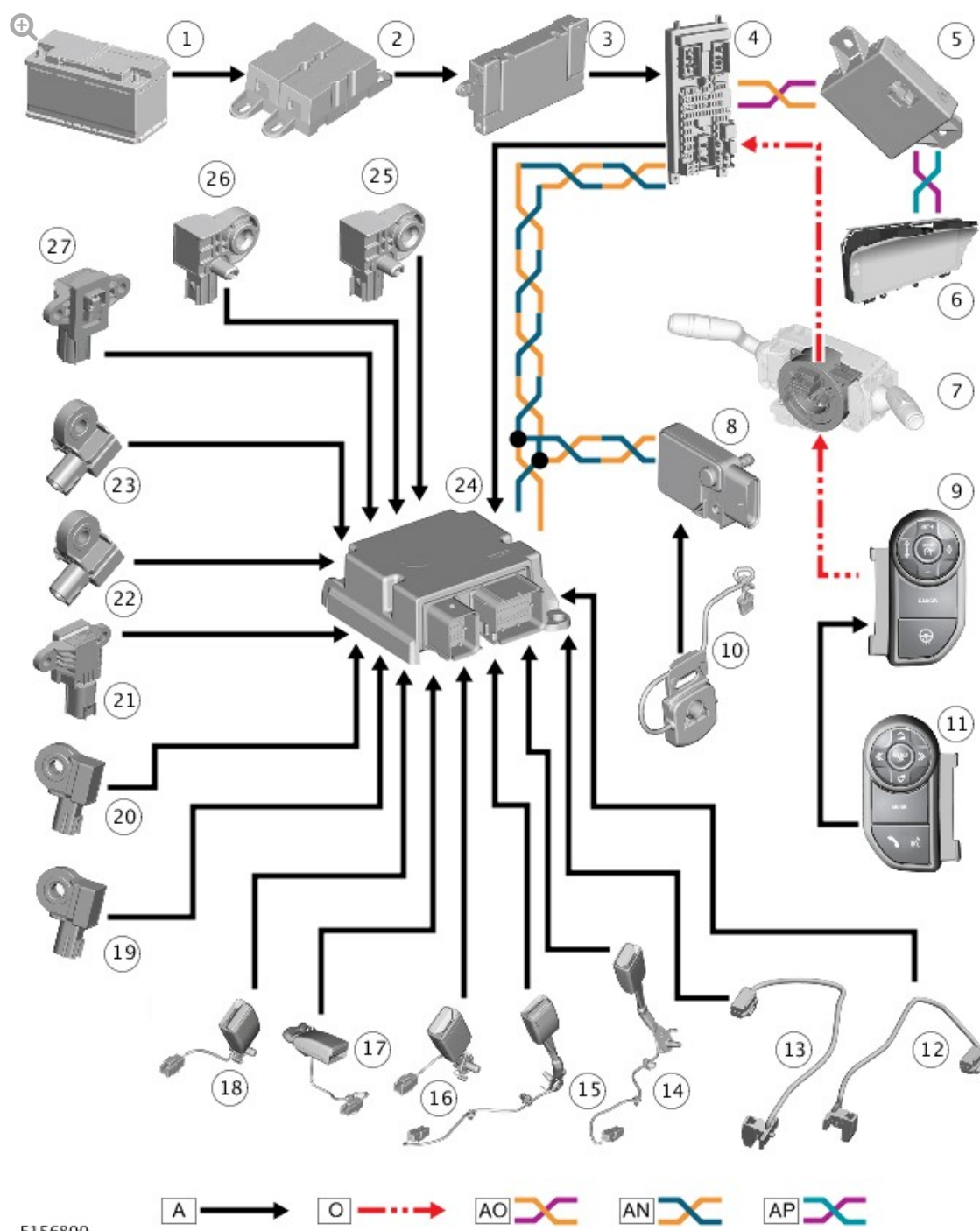
When the RCM outputs any of the fire signals it also outputs a crash signal to the CJB and the ECM (engine control module) on the high speed CAN powertrain systems. The crash signal is also hardwired to the ECM and the CJB. On receipt of the crash signal, the ECM cuts the power supply to the fuel pump relay and the CJB goes into crash mode. In the crash mode, the CJB:

- Activates all of the unlock signals of the vehicle locking system, even if the vehicle is already unlocked.
- Ignores all locking/superlocking inputs until it receives an unlock input, when it returns the locking system to normal operation.
- Activates the interior lamps. The interior lamps remain on permanently until they are manually switched off at the lamp unit, or the CJB crash mode is switched off and they return to normal operation.
- Disables the rear window child lock input until the crash mode is switched off.

- Activates the hazard flashers. The hazard flashers remain on until cancelled by the hazard warning switch or the crash mode is switched off.

The CJB crash mode is switched off by a valid locking and unlocking cycle of the locking system.

CONTROL DIAGRAM - SHEET 1 OF 2



A = HARDWIRED; O = LIN (LOCAL INTERCONNECT NETWORK) BUS; AO = MEDIUM SPEED CAN BODY SYSTEMS; AN = HIGH SPEED CAN

POWERTRAIN SYSTEMS; AP = MEDIUM SPEED CAN COMFORT AND CONVENIENCE SYSTEMS.

ITEM	DESCRIPTION
1	Battery
2	Battery junction box 2
3	Battery junction box
4	Central junction box
5	Gateway module
6	Instrument cluster and message center
7	Clockspring
8	Occupant classification sensor control module (NAS only)
9	Right steering wheel switchpack
10	Seatbelt tension sensor (NAS only)
11	Left steering wheel switchpack
12	Driver seat position sensor
13	Passenger seat position sensor (NAS only)
14	Driver seatbelt buckle switch
15	Passenger seatbelt buckle switch
16	Left rear seatbelt buckle switch
17	Center rear seatbelt buckle switch (where fitted)
18	Right rear seatbelt buckle switch
19	Left side impact sensor
20	Left side impact sensor
21	Left side pressure sensor
22	Front impact sensor
23	Front impact sensor
24	Restraints control module
25	Right side pressure sensor

6	Instrument cluster
7	Passenger airbag
8	Clockspring
9	Driver airbag
10	Passenger airbag deactivation indicator
11	Left rear retractor pretensioner
12	Right rear retractor pretensioner
13	Driver seatbelt pretensioner (where fitted)
14	Passenger retractor pretensioner (NAS non-active seatbelt version shown)
15	Driver retractor pretensioner (non-active seatbelt version shown)
16	Driver side airbag
17	Passenger side airbag
18	Restraints control module
19	Left side air curtain
20	Right side air curtain